

Course Code: CS1831		
Name of the Programme: BCA (Hons.)		
Name of the Course: Elements of Linear Algebra		
Semester: I		
Course Credit	No. of Hours Per Week (L + T + P)	Total No. of Teaching Hours
2	1:1:0	30
Course Objectives:		
a) To introduce the basic concepts of vectors and vector algebra for solving geometric and physical problems.		
b) To provide a foundational understanding of matrices and determinants and their applications in linear systems.		
c) To enable students to solve systems of linear equations using matrix-based methods such as Gauss Elimination and Inverse method.		
d) To familiarize students with linear transformations, eigenvalues, and eigenvectors for analyzing linear mappings and systems.		

Course Outline (Syllabus of the Course)

Module – 1 : Vector Algebra	No. of Hours
	3 Hrs.
Basics of vector, vector algebra, coplanarity, dot product, cross product, Linear combination of vectors, Linear dependence, Linear independence.	
Module – 2 : Matrices and Determinants	No. of Hours
	3 Hrs.
Matrix, matrix algebra, minor, cofactor, matrix transpose, adjoint matrix, special matrices, determinant, algebra of determinant	
Module – 3 : Rank and System of Linear Equations	No. of Hours
	3 Hrs.
Row Echelon form, Rank of a matrix. Solving system of linear equations using various methods: Inverse method, Gauss Elimination, Row canonical form, Gauss Jordan Method	
Module – 4 : Linear Transformation	No. of Hours
	3 Hrs.
Linear transformation, Properties of linear transformation, Rank, nullity.	
Module – 5 : Eigenvalues and Eigenvectors	No. of Hours
	3 Hrs.
Eigenvalues and eigenvectors, Characteristic equation, Diagonalizability, Geometric interpretation and application.	

Course outcomes

- a) Apply vector algebra concepts to solve mathematical and engineering problems effectively.
- b) Apply matrix algebra and determinants to solve complex mathematical problems.
- c) Solve systems of linear equations using various methods.
- d) Solve problems involving calculation related to eigenvalue and eigenvector for a given matrix.

Text Books

- 1 Rao, G. S. (2020). *Linear Algebra*. ISBN 978-93-89583-30-4.
- 2 Grewal, B. S. *Higher Engineering Mathematics* (ISBN 978-81-933284-9-1).
- 3 Strang, G. (2005). *Linear Algebra and Its Applications*. Pearson Education. ISBN 978-8131501726.

Reference books

- 1 Lipschutz, S., & Lipson, M. (1989). *3000 solved problems in linear algebra*. McGraw-Hill. ISBN: 0070380236
- 2 Lay, D. C., Lay, S. R., & McDonald, J. J. (2023). *Linear algebra and its applications* (5th ed.). Pearson. ISBN: 978-9357059688